TAHA ERKİN TUNALI

Mechanical Engineer

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Profile

I am a mechanical engineer with hands-on experience in engine performance, thermal management, and manufacturing optimization. Over the past years, I have worked on projects that range from improving production efficiency at Siemens to developing advanced simulation models for Ford's electric and combustion vehicles. I enjoy combining technical expertise with problem-solving skills to create practical and sustainable engineering solutions.

My background also includes academic research and a publication, which reflects my interest in applying innovative methods such as machine learning to automotive engineering challenges. I am motivated by opportunities where I can contribute to both technological development and environmental sustainability.

Education

2018 — **TED College High School, Konya**
GPA 3.55 / 4.00

2018 – 2023 — **Yildiz Technical University, Istanbul**
Department of Mechanical Engineering
GPA 3.07 / 4.00

Skills

CAD/CAE and Technical Drawing Tools

- AutoCAD
- SolidWorks
- Siemens NX
- CATIA
- Fusion360

Office and Data Visualization Tools

- Microsoft Office (Word, Excel, PowerPoint)
- Tableau

PLC and Automation Software

- Control FPWIN Pro

Programming and Computational Languages

- Python
- C

Simulation and Modeling Software

- GT-Suite
- MATLAB
- Ansys
- PVsyst
- RetScreen

Languages

English — Advanced (C1) | **German** — Elementary (A2) | **Romanian** — Beginner (A1)

Internship

06/2021 – 08/2021 **Akim Metal San. ve Tic. A.Ş.**
Manufacturing Internship

11/2022 – 01/2023 **Siemens San. ve Tic. A.Ş.**
Construction Internship

03/2023 – 05/2023 **Siemens San. ve Tic. A.Ş.**
Business and Organizational Internship

Professional Experience

09/2022 – 08/2023 **Part Time Working Student - Manufacturing Engineer**

Siemens San. ve Tic. A.Ş.

- Analyzed and visualized large-scale production data using **Excel**, **Python**, and **Tableau**, identifying process inefficiencies and enabling **~7% reduction** in cycle times
- Supported process management and project tracking across **5+ ongoing manufacturing improvement projects**, using **Microsoft Office** tools and **SharePoint**
- Performed root cause analyses for quality issues, improving defect detection rates by **~15%**
- Managed the coordination of design and order processes for production tools and components, ensuring **on-time delivery** across departments
- Developed image processing algorithms using **Python** and **OpenCV** to enhance visual inspection accuracy in quality control
- Actively contributed to digital transformation initiatives by integrating **data-driven dashboards** and automating manual reporting workflows
- Created detailed process flowcharts and documentation to support lean manufacturing and continuous improvement practices

01/2024 – Present **Propulsion Performance and Energy Management Engineer**

Ford Otomotiv Sanayi A.Ş.

- Utilized **CAE** tools for virtual validation of engine and propulsion systems, developed and optimized **simulation models** for motor performance, battery systems, and thermal management systems using **GT-Suite** and supported component selection.
- Prepared technical documentations and reports for **FO & European based projects** using **Powerpoint**, **Excel** and **Word**.

- Provided calibration support for both electric vehicles (EV) and internal combustion engines (ICE) under high-altitude and extreme ambient conditions using **MATLAB** and **GT-Suite**.
- Performed **thermal management** simulations for over 15 different vehicle variants, predicting flow rates, pressures and boost temperatures using **GT-Suite** to enhance performance and reliability.
- Executed friction and lubrication system analyses using **GT-Suite**, reducing mechanical losses by 3% through data driven optimization.
- Conducted hardware selection and **benchmarking** for heavy-duty trucks using **GT-Suite** and **Excel**, ensuring alignment with performance, emissions, and regulatory compliance targets.
- Led transmission efficiency evaluations and testing using **GT-Suite** and **Excel**, delivering gear-level efficiency improvement.
- Developed real-time simulation models (HiL, SiL, MiL) using **GT-Suite**, **Simulink**, and **Python**, reducing calibration time and effort by approximately **50%**.
- Developed an **AI-powered** knowledge assistant that retrieves accurate, context-based information from SharePoint documents. This agent significantly reduces search time and ensures procedural compliance by providing users with concise, cited answers from the most up-to-date sources.
- Designed, developed, and deployed a comprehensive **Python** tool to automate and accelerate key engineering tasks for the team. The tool significantly **reducing manual effort** and increasing productivity.

Graduate Thesis

02/2023 – 06/2023

BTruck (Hybrid Battery Electric / Fuel Cell Electric) and Investigation of Hydrogen Production Stations

- Conducted research on **sustainable** heavy-duty transportation **solutions** by integrating battery-electric and hydrogen fuel cell systems in hybrid trucks.
- Designed and analyzed **hydrogen** production stations powered by **renewable energy** to ensure eco-friendly fuel supply.
- Evaluated advantages and limitations of battery vs. fuel cell technologies, proposing a **hybrid** model to overcome range and charging challenges in long-distance freight transport.

Publications

10/2025

International Journal of Heat and Mass Transfer, *Range improvement in BEV trucks using heat pump for cabin heating instead of PTC heater*

09/2025

Thermal Science and Engineering Progress, *Machine Learning Based Predictive Control of Thermal Management System in Battery Electric Vehicles*

Projects

2019 – 2020	**Mechanical Pencil (Technical Drawing Project)**
2021	**A New Bicycle Conceptual Design with Systematic Design Approach**
2022	**Solar Energy Design and Analysis Konya**

Activies

2020 – 2021	**Formula Student Suspension Support Team Member**
2022	**Leap Event Case Competition Winner**